Dietary citric acid enhances absorption of aluminum in antacids.

Ten healthy men ingested, twice daily between meals, during each of the seven-day experimental periods: (a) citric acid (as lemon juice), (b) $Al(OH)_3$, or (c) $Al(OH)_3$ + citric acid. Whole blood sampled after each dietary period was analyzed electrothermally after digestion with nitric acid. Moderate, but significant, increases in mean Al concentrations as compared with pretreatment values [5 (SD 3) micrograms of Al per liter] were seen after ingestion of either citric acid or $Al(OH)_3$: 9 (SD 4) and 12 (SD 3) micrograms/L, respectively. Ingestion of both $Al(OH)_3$ and citric acid resulted in a more pronounced, highly significant (p less than 0.001) increase in Al concentrations, to 23 (SD 2) micrograms Al/L, probably owing to formation and absorption of Al-citrate complexes.